months or, conversely, capable of destroying in a second hundreds of thousands of human beings, in a city such as Hiroshima.

The theoretical speculations of the young Einstein had transported humanity into a new era: the era of nuclear power, an era of new possibilities, and new dangers. Today, thanks to the intelligence of a rebellious young man who would not abide rules, we have the instruments to bring light to the homes of the 10 billion human beings who will soon inhabit the planet, to travel in space towards other stars, or to destroy each other and devastate the planet. It depends on our choices; on which leaders we call upon to decide for us.

Today the structure of spacetime proposed by Einstein is well understood and repeatedly tested in laboratories; it is considered conclusively established. Time and space are different from the way they had been conceived since Newton. Space does not exist independently from time. In the expanded space of figure 3.2 there is no particular slice having a better claim than others to be called ‘space now’. Our intuitive idea of the present – the ensemble of all events happening ‘now’ in the universe – is an effect of our blindness: our inability to recognize small temporal intervals. It is an illegitimate extrapolation from our parochial experience.

The present is like the flatness of the Earth: an illusion. We imagined a flat Earth because of the limitations of our senses, because we cannot see much beyond our own noses. Had we lived on an asteroid of a few kilometres in diameter, like the Little Prince, we would have easily realized we were on a sphere. Had our brain and our senses been more precise, had we easily perceived time in nanoseconds, we would never have made up the idea of a ‘present’ extending everywhere. We would have easily recognized the existence of the intermediate zone between past and future. We would have realized that saying ‘here and now’ makes sense, but that saying ‘now’ to designate events ‘happening now’ throughout the universe makes no sense. It is like asking whether our galaxy is ‘above or below’ the galaxy of Andromeda: a question that makes no sense, because ‘above’ or ‘below’ has meaning on the surface of the Earth, not in the universe. There isn’t an ‘up’ or a ‘down’ in the universe. Similarly, there isn’t either always a ‘before’ and an ‘after’ between two events in the universe. The resulting knitted structure that space and time form together, depicted in figures like 3.2 and 3.3, is what physicists call ‘spacetime’ (figure 3.4).

When the *Annalen der Physik* published the article by Einstein in which all this was suddenly clarified, the impact upon the world of physics was momentous. The apparent contradiction between the equations of Maxwell and Newtonian physics were well known, and no one knew how to resolve them. Einstein’s solution, astonishing and extremely elegant, took everyone by surprise. The story goes that in the dimly lit old halls of Cracow